

Appendix B — At-a-glance KPI summary

(traceability matrix snapshot: 2025-12-29)

1 Verification and pipeline health

KPI	Readout	Status	Evidence ID
Repo unit tests + coverage	847/851 passed; 0 failed; 4 skipped; min coverage 71.98% (threshold 70%)	Met	GATE-UT
Workspace system suite	4/4 runs succeeded	Met	WS-01
Workspace ADQD validation suite	40/40 runs succeeded	Met	WS-02
Workspace sensitivity suite	240/240 runs succeeded	Met	WS-03
Workspace uncertainty suite	82/82 runs succeeded	Met	WS-04
Sensitivity “standard metrics” summary	missing	Not evaluated	MET-SA-QC
Uncertainty key-metrics summary	2031 interval new area (km²): base median 159.462; random median 137.800 (−13.6%); site-selection median 137.014 (−14.1%); base CV 0.014; base IQR 4.324.	Met	MET-UA-KEY

2 Benchmark headline metrics — Standard

Metric	Verification	Hold-out	Δ (Hold-out – Verification)
Change prevalence (observed)	1.12%	0.83%	-0.29 pp
Change prevalence (simulated)	0.55%	0.58%	+0.03 pp
Sim/Obs prevalence ratio	0.491×	0.704×	+0.213×
F1 (change mask)	0.354	0.166	-0.188
Quantity disagreement QD_change	0.475	0.313	-0.162
Allocation disagreement AD_change	0.334	0.621	+0.288
Spearman ρ (10 km)	0.583	0.515	-0.068

3 Benchmark headline metrics — influence-zone 500m

Metric	Verification	Hold-out	Δ (Hold-out – Verification)
Change prevalence (observed)	3.24%	3.72%	+0.48 pp
Change prevalence (simulated)	2.52%	2.56%	+0.04 pp
Sim/Obs prevalence ratio	0.777×	0.688×	-0.088×
F1 (change mask)	0.169	0.134	-0.036
Quantity disagreement QD_change	0.300	0.348	+0.048
Allocation disagreement AD_change	0.666	0.637	-0.029
Spearman ρ (10 km)	0.526	0.467	-0.059

Notes:

- Δ is computed as (Hold-out – Verification). “pp” = percentage points.
- For interpretation: higher is better for F1/ ρ ; lower is better for QD/AD.